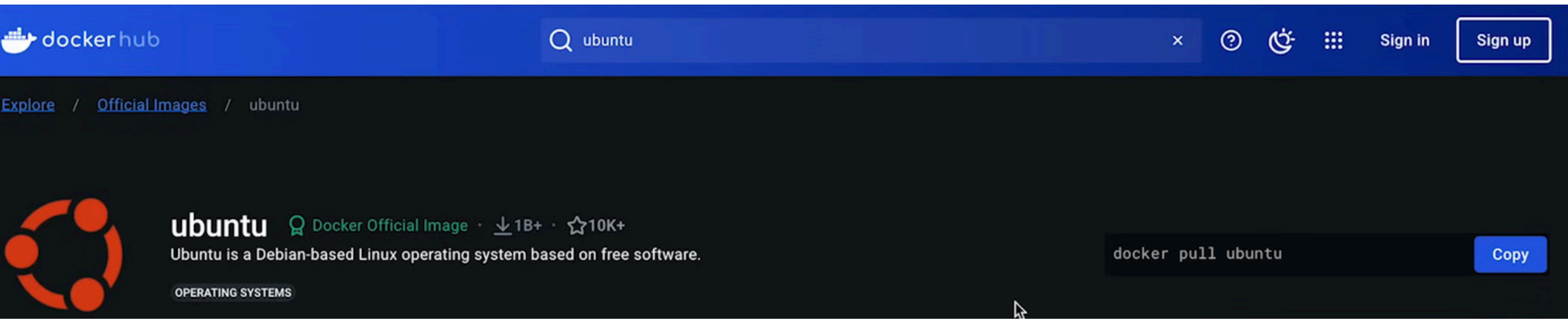
Docker #Day 1

Run Ubuntu in Interactive mode



Command: docker run -it ubuntu

It downloaded ubuntu image from Ubuntu Hub

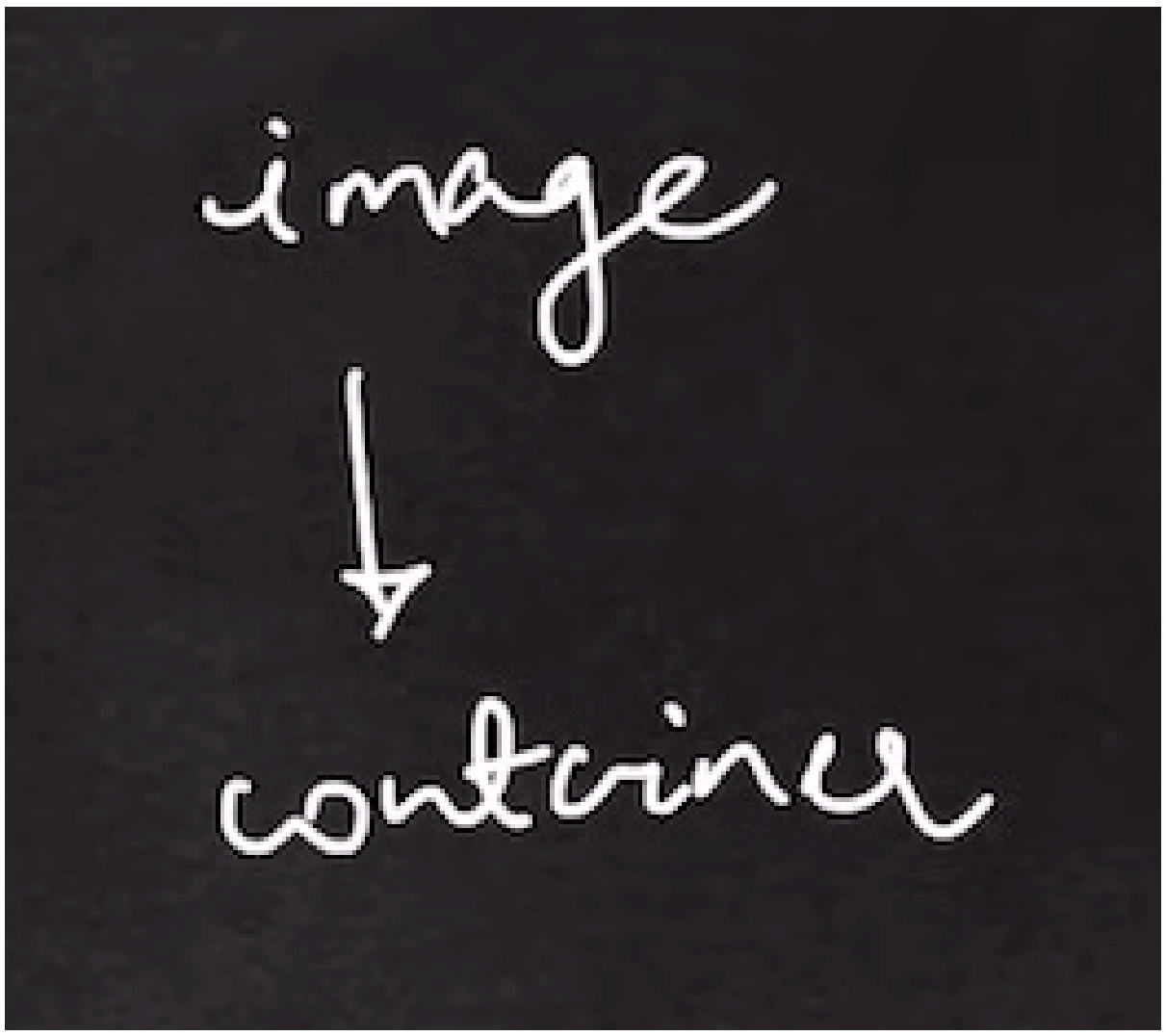
```
root@ff4e5c0c0895: /
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Muhammad Bilal>docker run -it ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
92842f25412d: Download complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b
Status: Downloaded newer image for ubuntu:latest
root@ff4e5c0c0895:/#
```

First we were in Windows terminal but now in Root Directory in Ubuntu Container

```
root@ff4e5c0c0895:/#
```

Now we can add or delete and perform operations and it will not effect our windows operating system

Now to create Docker images and Containers



The Rule is simple:

1) create docker image first (we use docker pull command)

2)Then create docker container (we use docker run command)

Command: Docker pull hello-world

Administrator: Command Prompt

```
C:\Users\Muhammad Bilal>docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7da
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest

What's next:
    View a summary of image vulnerabilities and recommendations → d
```

Command: **Docker run hello-world**

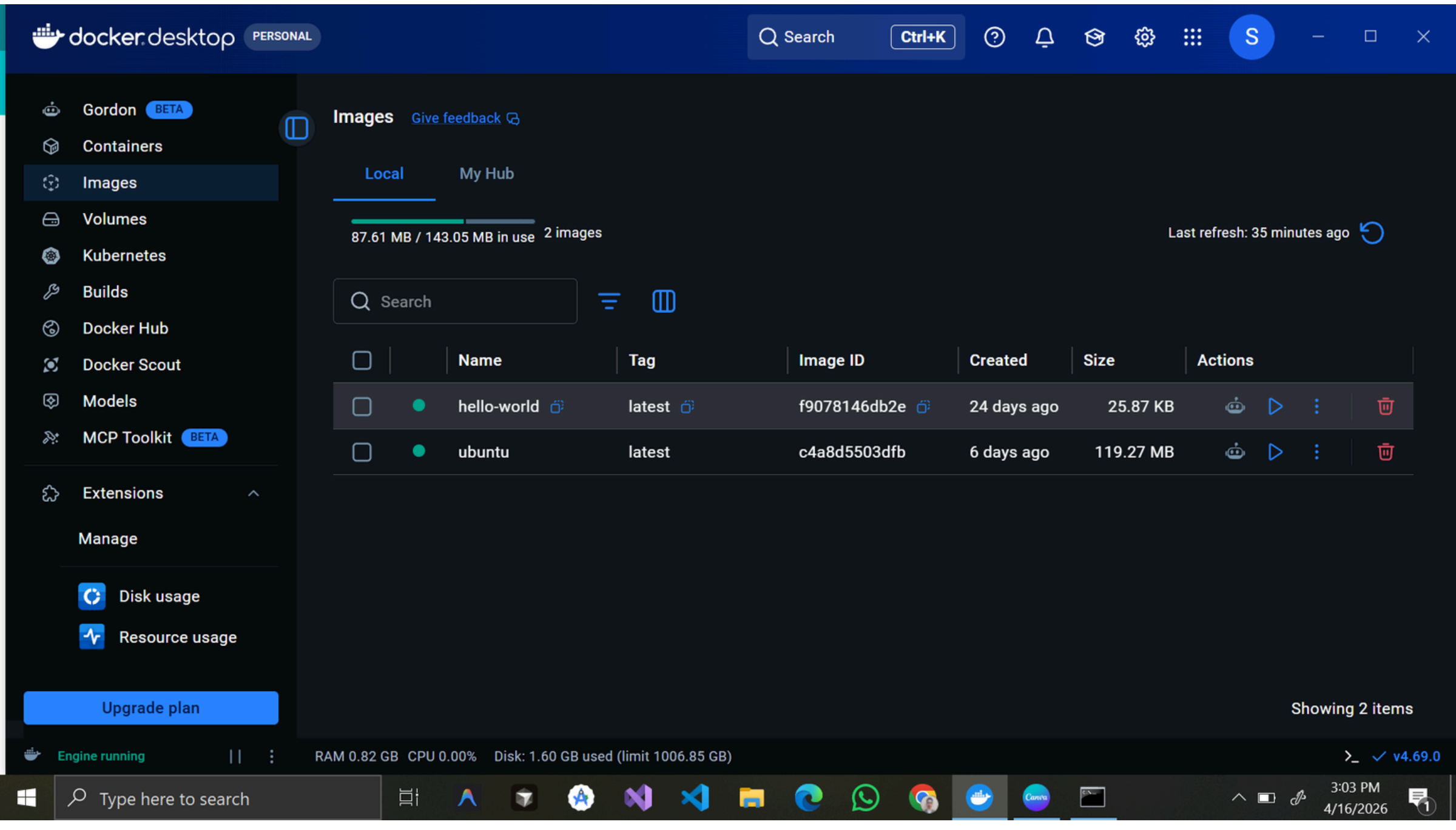
```
C:\Users\Muhammad Bilal>docker run hello-world

Hello from Docker!
This message shows that your installation appears to be workin

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the
    (amd64)
 3. The Docker daemon created a new container from that image
    executable that produces the output you are currently read
 4. The Docker daemon streamed that output to the Docker clien
    to your terminal.

To try something more ambitious, you can run an Ubuntu contain
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

First How to check how many images we have currently in Docker?

Command: docker images

```
C:\Users\Muhammad Bilal>docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     f9078146db2e       25.9kB         9.49kB    U
ubuntu:latest          c4a8d5503dfb        119MB         31.7MB    U
```

you can see in docker destop to verify too



Command: ls

(ls means list command)

```
root@90735c8a1db9:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt
root@90735c8a1db9:/#
```

Command: mkdir bilal

mkdir mens (make directory) and then you can create directory of your own name

```
root@90735c8a1db9:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt
root@90735c8a1db9:/# mkdir bilal
root@90735c8a1db9:/# ls
bilal  bin  boot  dev  etc  home  lib  lib64  medi
root@90735c8a1db9:/#
```

Command: env

```
root@90735c8a1db9:/# env
HOSTNAME=90735c8a1db9
PWD=/
HOME=/root
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;
30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.
gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01
=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;
=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gi
.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.p
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*
;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35
;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*
5:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:
crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=
;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
*.ucf-old=00;90:
TERM=xterm
SHLVL=1
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/b
_=/usr/bin/env
root@90735c8a1db9:/#
```

It Prints the environment vaiables

If you want to go back from root directory to windows terminal

Command: exit

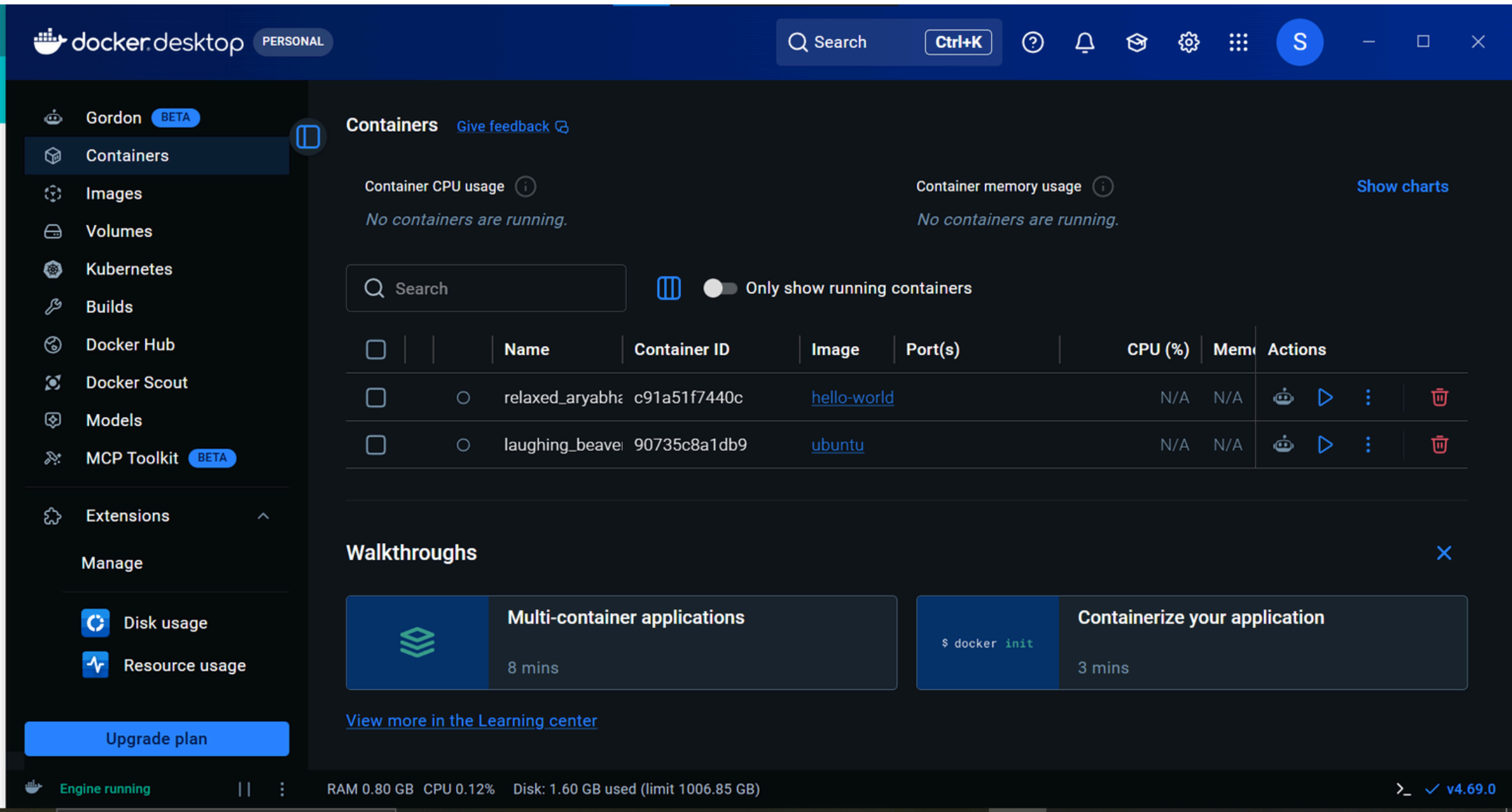
```
root@90735c8a1db9:/# exit
exit

C:\Users\Muhammad Bilal>
```

Check all the current containers in Docker

Command: docker ps -a

```
C:\Users\Muhammad Bilal>docker ps -a
CONTAINER ID   IMAGE          COMMAND        CREATED          STATUS
90735c8a1db9   ubuntu         "/bin/bash"    5 minutes ago    Exited (0) Abo
c91a51f7440c   hello-world    "/hello"       10 minutes ago   Exited (0) 10
```

You can also check containers in docker app



check running status of any container

Command: docker ps

```
C:\Users\Muhammad Bilal>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS
```

Now you can see that there are no current running containers

Command: **docker start (**add container id or name**)**

eg: first copy container id in Docker image

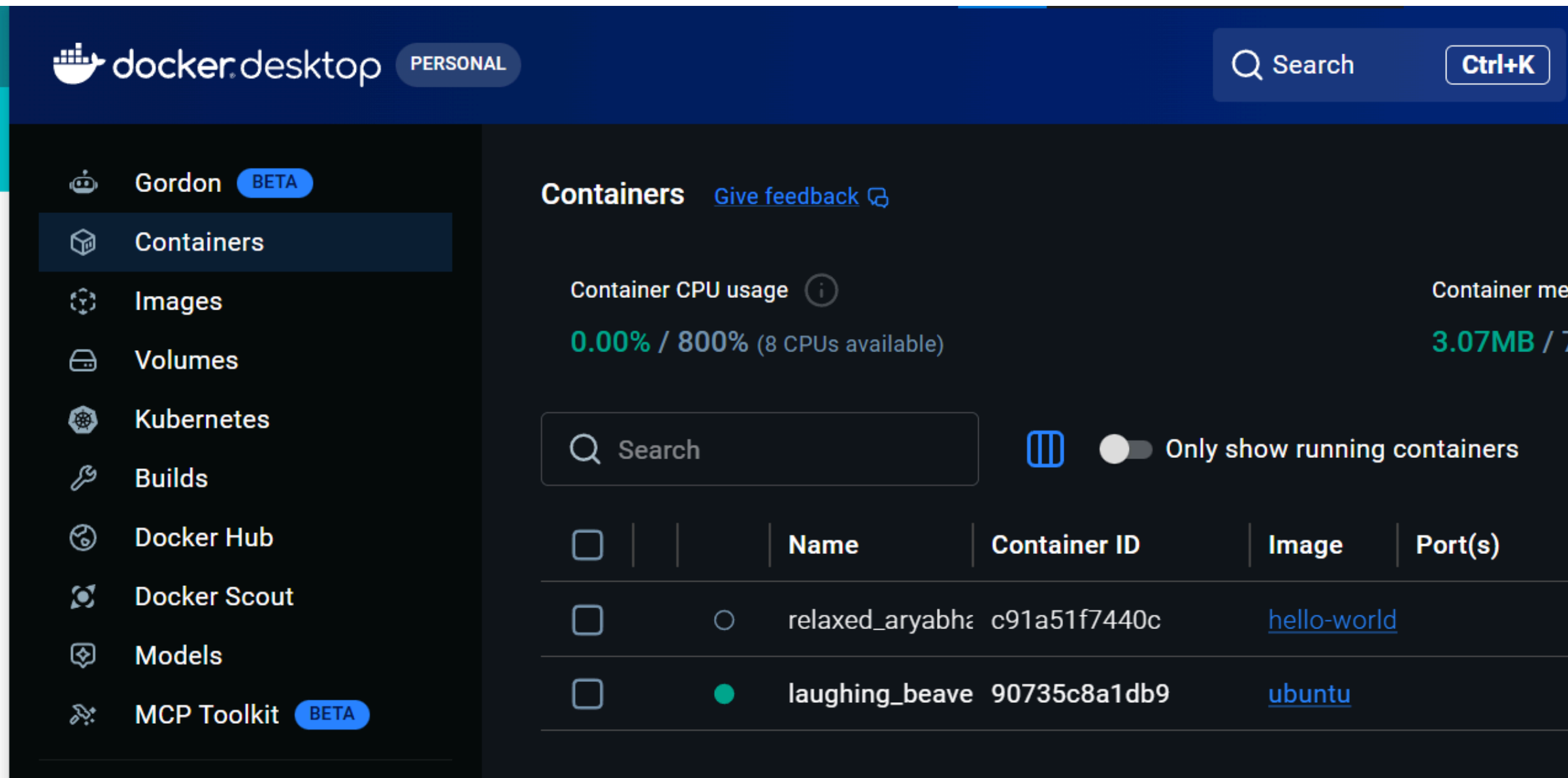
Container ID	Image	Port(s)
c91a51f7440c	hello-world	
90735c8a1db9	ubuntu	

```
C:\Users\Muhammad Bilal>docker start 90735c8a1db9f102d3
90735c8a1db9f102d32a9620c06283e491121a757687f5ffd3be796
```

now see current running contianer through docker ps command

```
C:\Users\Muhammad Bilal>docker start 90735c8a1db9f102d32a9620c06283e4
90735c8a1db9f102d32a9620c06283e491121a757687f5ffd3be796ddb50108b

C:\Users\Muhammad Bilal>docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS
90735c8a1db9   ubuntu    "/bin/bash"   12 minutes ago   Up 43 seconds
```

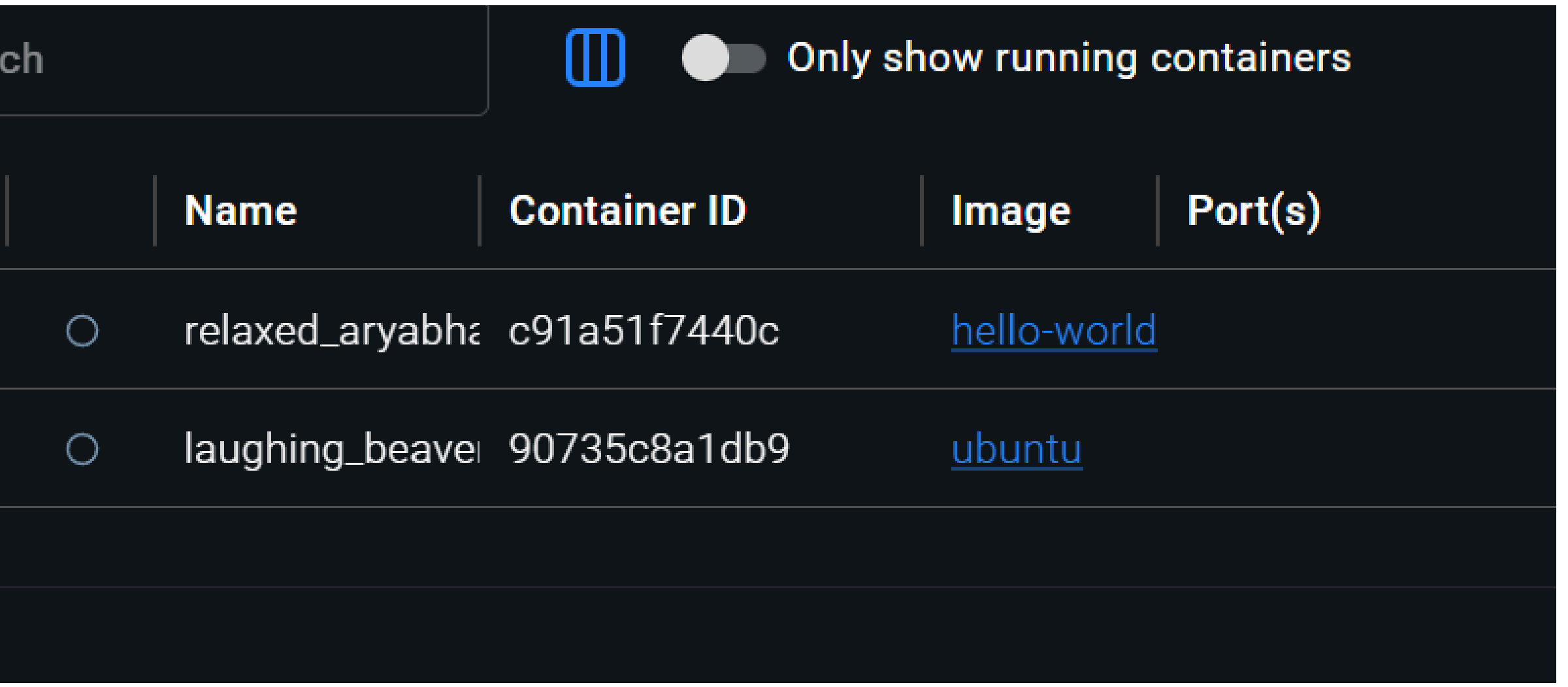
See the green dot in the ubuntu container means its running



Command: **Docker stop (add the name or id)**

```
C:\Users\Muhammad Bilal>docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS
90735c8a1db9   ubuntu    "/bin/bash"   12 minutes ago   Up 43

C:\Users\Muhammad Bilal>docker stop  laughing_beaver
laughing_beaver
```

I have used docker image name but you can use the docekr id to stop



see green dot in conainer in app its gone menas it has been stopped

Now to delete container first we have to delete image then we can delete its container

<u>Note:</u> if we directly try to delete container it will give error

Command: docker rmi (add container id or name with it)

```
C:\Users\Muhammad Bilal>docker images

IMAGE                    ID              DISK USAGE
hello-world:latest       f9078146db2e       25.9kB
ubuntu:latest            c4a8d5503dfb        119MB

C:\Users\Muhammad Bilal>docker rmi  c4a8d5503df
Error response from daemon: conflict: unable to
ntainer 90735c8a1db9
```

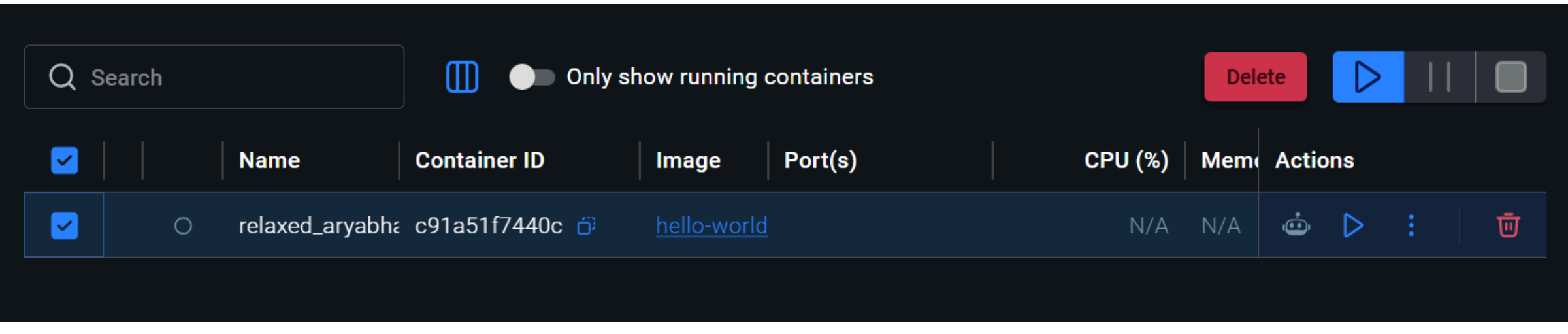
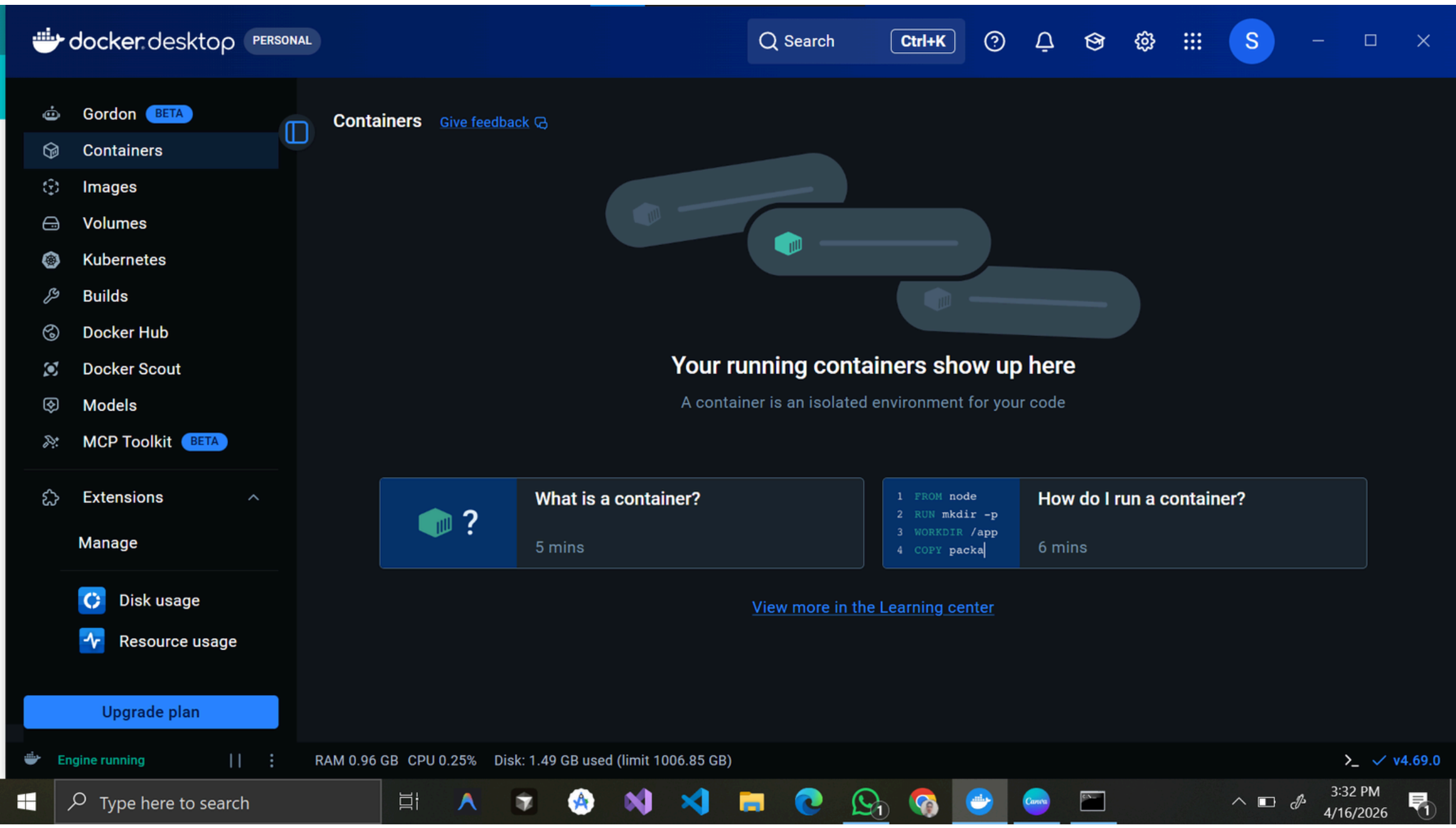
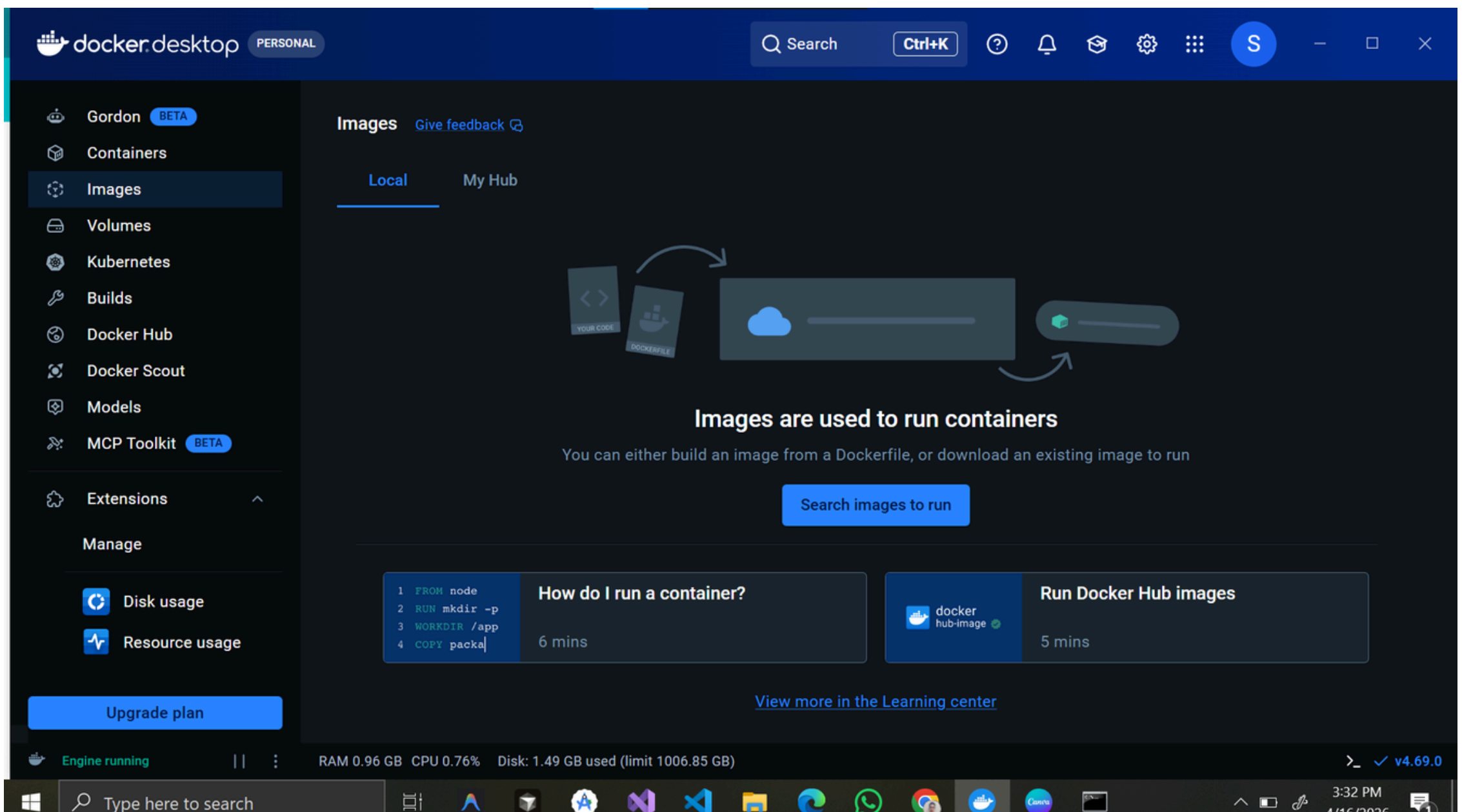
As previously we have deleted the docker image now we can easily delete the container first

Command: docker rm (add docker container id or name)

```
C:\Users\Muhammad Bilal>docker ps -a
CONTAINER ID   IMAGE          COMMAND       CREATED
90735c8a1db9   ubuntu         "/bin/bash"   22 minutes ago
c91a51f7440c   hello-world    "/hello"      28 minutes ago

C:\Users\Muhammad Bilal>docker rm 90735c8a1db9
90735c8a1db9
```

It will return see the specific container of ubuntu has been finally deleted

we can also delete it through simply by click



now you can see no images or containers in the Dokcer after deleting all the images and containers